

3.8.1 Centre Assessed

Route T (PSA/ISA)

Each centre assessed unit comprises

- Practical Skills Assessment (PSA)
- Investigative Skills Assignment (ISA).

The PSA consists of the centre's assessment of the candidate's ability to demonstrate practical skills throughout the course; thus, candidates should be encouraged to carry out practical and investigative work throughout the course of their study. This work should cover the skills and knowledge of *How Science Works* (Section 3.7) and in Sections 3.3 and 3.6.

The ISA has two stages where candidates

- undertake practical work, collect and process data
- complete a written ISA test.

Each stage must be carried out under controlled conditions but may be scheduled at a time convenient to the centre. The written test must be completed in a single, uninterrupted session.

The ISA is set externally by AQA, but internally marked, with marking guidelines provided by AQA.

In a given academic year two ISAs at each of AS and A2 level will be provided.

inorganic, physical and organic chemistry.

AS level candidates should undertake at least **two of the tasks** from each of the three areas of chemistry. Each task will score a maximum of 2 marks aggregating to a total score out of 12 marks.

Only marks arising from those tasks designated as AS tasks may be submitted for AS Chemistry.

A2 level candidates should undertake at least **two of the tasks** from each of the three areas of chemistry. Each task will score a maximum of 2 marks aggregating to a total score out of 12 marks. Only marks arising from those tasks designated as A2 tasks may be submitted for A2 Chemistry.

At both AS and A2, centres may choose to carry out all of the available tasks, in which case, the **best two marks** from each area for each candidate should be counted towards the final mark.

A list of the AS and A2 PSA tasks is given in Sections 3.3 and 3.6, respectively. Detailed marking guidance including descriptors for 0, 1 and 2 marks for each PSA task is provided in Section 3.8.3.

Candidates should be awarded marks which reflect their level of performance over the whole course.

AQA may wish to ask for further supporting evidence from centres in relation to the marks awarded for the PSA. Centres should therefore keep records of their candidates' performances in their practical activities throughout the course. (For example, a laboratory

diary, log or tick sheet.)

Further guidance for the awarding of marks for the PSA will be provided in the Teacher Resource Bank.

Use of ICT during the PSA

Candidates are encouraged to use ICT where appropriate in the course of developing practical skills, for example in collecting and analysing data.

Investigative Skills Assignment (ISA)

The Investigative Skills Assignment carries 38 marks and has two stages.

Stage 1: Collection and Processing of data

Candidates carry out practical work following an AQA task sheet. Centres may use the task sheet as described or may make minor suitable modifications to materials or equipment, following AQA guidelines. Any modifications made to the task sheet must be agreed in writing with the AQA Assessment Adviser. The task may be conducted in a normal timetabled lesson but must be under controlled conditions.

Candidates collect raw data and represent it in a table of their own design or make observations that are recorded on the *Candidate Results Sheet*. The candidates' work must be handed to the teacher at the end of the session. The teacher assesses the candidates' work following AQA marking guidelines.

There is no specified time limit for this stage.

Stage 2: The ISA written test

The ISA test should be taken as soon as convenient after completion of Stage 1, and under controlled conditions. Each candidate is provided with an ISA test and the candidate's completed material from Stage 1. The teacher uses the AQA marking guidelines to assess the ISA test.

The ISA test is in two Sections.

Section A

This consists of a number of questions relating to the candidate's own data.

Section B

At the start of this section, candidates are supplied with additional data on a related topic. A number of questions relating to analysis and evaluation of the data then follow.

The number of marks allocated to each section may vary slightly with each ISA test.

Use of ICT during the ISA

ICT may be used during the ISA Stages 1 and 2 but teachers should note any restrictions in the ISA marking guidelines or Teachers Notes. Use of the internet is not permitted.

Candidates absent for the practical work

A candidate absent for the practical work (Stage 1) should be given an opportunity to carry out the practical work before they sit the ISA test. This may be with another group or at a different time. In extreme circumstances, when such arrangements are not possible, the teacher may supply a candidate with class data. In this case candidates cannot be awarded marks for Stage 1, but can still be awarded marks for Stage 2 of the assessment.

Material from AQA

For each ISA, AQA will provide:

- Teachers' Notes
- Task sheet
- ISA test
- Marking guidelines.

This material must be kept under secure conditions within the centre. If it is to be used in more than one session, then the centre must ensure security of the material between sessions. Further details regarding this material will be provided.

Security of assignments

Completed ISAs should be treated like examination papers and kept under secure conditions until the publication of results.

General Information Route T

Administration

In any year a candidate may attempt either or both of the two ISAs.

For each candidate, the teacher should submit to AQA a total mark comprising:

- The PSA mark
- the better ISA mark (if two have been attempted).

The ISA component of this mark must come from one ISA only, i.e. the marks awarded for stages of different ISAs cannot be combined.

Candidates may make only one attempt at a particular ISA. Redrafting is not permitted at any stage during the ISA.

The total mark must be submitted to AQA by the due date in the academic year for which the ISA was published.

Work to be submitted

For each candidate in the sample the following materials must be submitted to the moderator by the deadline issued by AQA.

- the candidate's data from Stage 1 (on the *Candidate Result Sheet*)
- the ISA written test, and
- a *Candidate Record Form* showing the marks for the ISA and the PSA.

In addition each centre must provide

- *Centre Declaration Sheet*
- Details of any amendments to the task sheet with information supporting the changes from the AQA Assessment Adviser, must be notified to the moderator.

Working in groups

For the PSA candidates may work in groups provided that any skills being assessed are the work of individual candidates. For the ISA further guidance will be provided in the Teacher Notes.

Other information

Section 6 of this specification outlines further guidance on the supervision and authentication of centre assessed units.

Section 6 also provides information in relation to the internal standardisation of marking for these units. Please note that the marking of both the PSA and the ISA must be internally standardised as stated in Section 6.4.

Further support

AQA supports the units in a number of ways.

- AQA holds annual standardising meetings on a regional basis for all internally assessed components. Section 6 of this specification provides further details about these meetings
- A Teacher Resource Bank which includes further information and guidance
- Assessment Advisers are appointed by AQA to provide advice on internally assessed units. Every centre is allocated an Assessment Adviser.

The Assessment Advisers can provide guidance on

- issues relating to the carrying out of tasks for assessment
- application of marking guidelines

Any amendments to the ISA task sheet must be discussed with the Assessment Adviser and confirmation of the amendments made must be submitted to the AQA moderator.

conditions but may be at a time convenient to the centre. The written test must be completed in a single uninterrupted session.

The EMPA is set and marked by AQA. Only one EMPA at AS and one at A2 will be provided in a given academic year. AQA will stipulate a period of time during which the EMPA (task and written test) must be completed.

Practical Skills Verification

Candidates following this route must undertake the practical activities outlined in sections 3.3 for AS or 3.6 for A2 in order to allow candidates suitable opportunities to demonstrate safe and skilful practical techniques and to make reliable and valid observations. The teacher will confirm on the *Candidate Record Form*, for each candidate that this requirement has been met. Failure to complete the tick box will lead to a mark of zero being awarded to the candidate for the whole of this unit. Knowledge and understanding of the skills in Section 3.3 and 3.6 will be assessed in Section C of the EMPA written tests.

Tasks are provided in the **three areas** of chemistry; inorganic, physical and organic chemistry. Candidates should undertake at least two of the task form from each of the three areas of chemistry.

In order to provide appropriate opportunities to demonstrate the necessary skills, teachers must not be too prescriptive to the instructions they provide but should allow candidates to make decisions for themselves, particularly concerning the conduct of practical work, their organisation and the manner in which equipment is used.

Candidates should be encouraged to carry out practical and investigative work throughout the course. This work should cover the skills and knowledge of *How Science Works* (Section 3.7) and in Sections 3.3 and 3.6. Alternative practical activities may be used provided that candidates are given the opportunity to develop the same skills identified in Section 3.3 or 3.6.